SeaArt: Making AI image generation accessible for everyone with cloud

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SeaArt deploys it
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About SeaArt

Founded in 2023, SeaArt is dedicated to lowering the technical barrier of leveraging the latest generative AI technologies for all. SeaArt AI, its AI image generation service platform, was launched in March 2023 and reached 20x user growth within five months. The platform currently has around 60,000 daily active users around the world.

Industries: Technology

Google Cloud results

- Reduces 30 percent of infrastructure costs by enhancing hardware utilization with Spot VMs and Cloud GPUs
- Enables one-person DevOps team with easy-to-manage, high-performance cloud infrastructure
- Enhances first-time web page loading speed by 90 percent globally through Cloud Load Balancing
- Supports real-time data analytics for timely troubleshooting with Firebase and BigQuery

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Location: Singapore

(https://cloud.google.com/load-balancing/),

Google Cloud (https://cloud.google.com/)

BigQuery (https://cloud.google.com/bigquery/)

Cloud CDN (https://cloud.google.com/cdn/)

Media CDN (https://cloud.google.com/media-cdn)

Cloud GPUs (https://cloud.google.com/gpu/)

Cloud Load Balancing (https://cloud.google.com/load-balancing/)

Cloud SDK (https://cloud.google.com/sdk/)

Cloud Storage (https://cloud.google.com/storage/)

Firebase (https://firebase.google.com/)

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Cloud Firestore (https://cloud.google.com/firestore/)

Google Cloud Armor (https://cloud.google.com/armor/)

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

Spot VMs (https://cloud.google.com/spot-vms)

Generative artificial intelligence (AI) has advanced in both technology maturity and popularity in 2023. A survey published in August 2023 shows (https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai-in-2023-generative-ais-breakout-year) that one-third of organizations are using generative AI tools regularly less than a year after many of these tools were released. However, not all generative AI tools are easy to use. For example, some AI image generation tools require complex configurations to deliver high-quality pictures.

Founded in early 2023, SeaArt (https://www.seaart.ai/home) is dedicated to lowering the technical barrier of leveraging the latest generative AI technologies for all users. The SeaArt team is experienced in optimizing generative AI tools. SeaArt AI, the company's AI image generation service platform, enables users to quickly create high-quality AI-generated images by automatically recommending the most suitable image generation AI models and configurations to use. Launched in March 2023, SeaArt AI reached 20 times user growth within five months and currently has around 60,000 daily active users around the world.

"Our mission is to make generative AI accessible for everyone," explains Fei Ma, CEO at SeaArt. "We built an automated recommendation system that can suggest the best AI models and configurations to use based on the prompts that users provide and the image style they prefer. This way, our users can create quality AI-generative images by one to two clicks, instead of navigating through complex configurations by themselves."

In the beginning of developing its image generation platform, SeaArt ran its machine learning (ML) models on a cloud platform that didn't support flexible computing resources management, which complicated the team's development process. At the same time, SeaArt wanted to leverage GPU spot instances to enhance the cost-effectiveness of its operations. The company then started deploying its image generation ML models in Google Cloud (https://cloud.google.com/), because it supports more efficient DevOps and provides GPU spot instances with the NVIDIA L4, one of the latest graphics cards released in March 2023.

"As a small team exploring new AI technologies like image generation, we want to keep our DevOps workloads as low as possible and concentrate on innovation," says Ma. "Google Cloud meets this need by providing high-performance infrastructure that enables efficient operations. Besides, it possesses rich, cost-effective computing resources with the most advanced hardware that can facilitate our AI services."

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Enabling cost-effective AI image generation with powerful cloud infrastructure

With Google Cloud, the SeaArt team swiftly built a fully-managed cloud environment hosting its ML models within one week. Since the beginning, the Google Cloud team has been offering a wide range of technical advice, such as machine type selection, architecture design, global resource distribution and network security, to facilitate SeaArt's adoption.

"We've been in close communication with the Google Cloud team since we adopted Google Cloud, and they're always keen to find the best solutions for our use case," notes Ma. "Thanks to the comprehensive advice that we've received from Google Cloud engineers, we're able to quickly build and deploy our AI image generation platform and continue improving our services."

SeaArt deploys the NVIDIA L4 GPUs in Cloud GPUs (https://cloud.google.com/gpu) in Spot VMs (https://cloud.google.com/spot-vms), running as managed instances, in Google Kubernetes Engine (https://cloud.google.com/kubernetes-engine) (GKE) for its AI image generation services, and stores user behavior data in Firestore (https://cloud.google.com/firestore) and application files in Cloud Storage (https://cloud.google.com/storage). Ma says that the high performance of these infrastructure tools has greatly benefited SeaArt's operations. With the NVIDIA L4 GPUs available as spot instances, the SeaArt team can not only run ML workloads in a more cost-effective manner, but also quickly launch and remove computing resources to enhance its hardware utilization and reduce 30 percent of infrastructure costs. Besides providing nearly unlimited scalability, Cloud Storage also supports simultaneous data storage in multiple regions. This way, SeaArt's users are able to access data through private networks in each region, which enhances network throughput capacity and lowers data transferring costs.

"By leveraging Cloud GPUs attached to Spot VMs in GKE, we're able to run our ML models with excellent performance, cost-effectiveness, and scalability," he adds. "Since its launch, our image generation platform has experienced 20 times user growth in just five months, and we've never encountered any scalability issues with the infrastructure tools of Google Cloud."

On top of that, SeaArt has greatly simplified its DevOps process by adopting Cloud SDK (https://cloud.google.com/sdk) to access and manage Google Cloud tools from its local machines. For example, the team can manage GKE instance groups from the command line, which is easier than doing so from the GKE interface. Through Cloud SDK, it can also automate Docker image building in Google Cloud to shorten new feature deployment time. As a result, SeaArt has been able to rely on only one engineer to cover all the DevOps work.

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Providing speedy content delivery with high network security

To offer quality AI image generation services around the world, SeaArt needs to ensure that its users can load images as fast as possible. To that end, it uses Cloud CDN and Media CDN (https://cloud.google.com/cdn) to deliver content worldwide, eventually reaching an image loading rate of 99.5 percent. In addition, the company leverages Cloud Load Balancing (https://cloud.google.com/load-balancing/) to accelerate web page loading speed by distributing network traffic among servers. Overall, the time to load SeaArt's web pages for first-time users is shortened by 90 percent.

In addition, cyberattacks are rampant in a new domain like generative AI services. Each time an attack took place in the past, the traffic to the organization's image generation platform increased drastically, which generated additional network costs. To reinforce its network security, SeaArt employs Google Cloud Armor (https://cloud.google.com/armor/) to mitigate web attack risks. Since the adoption, its platform has not been impacted by cyberattacks.

Supporting timely troubleshooting and data-driven decision making with BigQuery

SeaArt has built a data-driven culture and relies on data analytics to continue improving its operations and products, with BigQuery (https://cloud.google.com/bigquery/) as its data warehouse. Ma notes that since SeaArt's platform generates three million images every day, the high performance and scalability of BigQuery have been very helpful in processing a large amount of data. The team has also optimized queries in BigQuery through data tagging.

To troubleshoot more quickly, SeaArt has also been analyzing data in real time by sending its image generation platform's user behavior data through Firebase (https://firebase.google.com/) to BigQuery. When a technical issue is detected through analytics, an alert will be immediately sent to the SeaArt team, so that it can react to ensure its service quality in a timely manner. For example, when the login success rate of its platform drops to lower than 98 percent, the team will receive an alert and instantly start to identify the cause.

"We use data analytics results as guidance to our business decisions in all respects from product development to user retention. BigQuery enables us to easily process ever-increasing data and realize real-time analytics to improve our services effectively and efficiently," says Ma.

Boosting business growth and technology advancement with Google Cloud

With the highly scalable infrastructure of Google Cloud, SeaArt is prepared to continue acquiring new users at a rapid pace. Next, it plans to advance its generative AI technology to support the generation of moving images, short and long animations using the computing resources in Google Cloud.

Ma says, "We're confident that the AI image generation services we've built on Google Cloud are scalable and powerful enough to support continuous exponential growth. We'll keep developing new and better generative AI products leveraging the top-notch computing capabilities of Google Cloud."

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